

TailWatch: An InSAR Deformation-Monitoring and Inverse-Velocity Failure-Forecasting Studio, on Synthetic Sentinel-1 and Real Campi Flegrei

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 Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_TailWatch

Abstract. Ground deformation is the earliest measurable warning of a tailings-dam or pit-wall failure, and satellite interferometric synthetic-aperture radar (InSAR) can measure it at the millimetre scale over a whole facility. This report describes TailWatch, an interactive, reproducible studio that carries a deformation field from InSAR displacement to a failure forecast, and measures the accuracy of each stage. Two data lanes feed it: a high-fidelity synthetic Sentinel-1 forward simulation (von-Kármán atmospheric phase screen, decorrelation, digital-elevation-model error and orbital ramps) over five deformation regimes, and one *real* case, the Campi Flegrei caldera from the open COMET LiCSAR/LiCSBAS Sentinel-1 archive. A two-geometry Up/East decomposition resolves the synthetic line-of-sight field; a learned tier flags anomalies and classifies deformation type; and a classical inverse-velocity model forecasts the time of failure. All accuracies are measured on synthetic data. On held-out synthetic scenes the anomaly detectors are strong (velocity-based area-under-curve 0.968, autoencoder 0.898), and while the six-class deformation-type classifier is only moderate overall (macro-F1 0.556), it recovers the one class that matters for early warning, *accelerating*, at 0.99 recall. On 40 synthetic accelerating trajectories the inverse-velocity forecast detects every failure (100%) with a median failure-time error of 5.7%, tightening from about 14% far ahead to 2% close to failure, the behaviour a warning system needs, and it raises no false alarm on 60 non-accelerating synthetic controls. The Campi Flegrei series (2016–2018) contains no failure event, so no failure-time error can be measured on it; the studio computes its line-of-sight velocity, coherence and cumulative displacement, and runs the synthetic-trained detectors on it without ground truth. TailWatch is a didactic and decision-support instrument, not a certified safety system, and the small-baseline network inversion is not implemented. Every number and figure can be regenerated from the released artifacts.

Index Terms. InSAR, ground deformation, tailings dam, slope stability, inverse velocity, failure forecasting, anomaly detection, Sentinel-1, Campi Flegrei, reproducibility.

I. INTRODUCTION

THE failure of a tailings storage facility or a pit wall is preceded, often by weeks, by accelerating ground

deformation, and measuring that deformation early is the core of geotechnical risk management. Satellite interferometric synthetic-aperture radar (InSAR) is uniquely suited to it: by comparing the phase of radar returns across repeated passes it measures line-of-sight displacement at the millimetre scale over an entire facility, without instrumenting the ground [1], [2]. Two families of processing make the measurement robust to noise, persistent-scatterer interferometry [3], [5] and small-baseline subset (SBAS) time-series analysis [4], and the open Sentinel-1 archive, processed by automated pipelines such as LiCSAR and LiCSBAS [6], [7], has made global monitoring routine. What a mine operator needs from that displacement field is not the field itself but a decision: is anything anomalous, what kind of movement is it, and, if it is accelerating, when will it fail?

TailWatch is an interactive studio that makes that chain explicit; this report describes it and measures the accuracy of each stage. It is built on two data lanes (Fig. 1). The first is a high-fidelity synthetic Sentinel-1 forward simulation, which lets the pipeline be exercised over controlled deformation regimes with known ground truth; the second is a single *real* case, the Campi Flegrei caldera near Pozzuoli, drawn from the open COMET LiCSAR/LiCSBAS Sentinel-1 archive, on which the same pipeline processes real Sentinel-1 motion. From the displacement field a two-geometry decomposition resolves vertical and east-west motion (vertical only on the single-geometry real frame), a learned tier flags spatial anomalies and classifies each pixel’s deformation type, and a classical inverse-velocity model projects the failure time with a tiered alarm. The contribution is not a new InSAR algorithm; it is a reproducible, end-to-end studio with the accuracy of each stage measured on synthetic data, including where it is strong and where it is only moderate.

II. THE PIPELINE AND ITS DATA

The synthetic lane generates a Sentinel-1 line-of-sight interferometric stack by a forward model that adds, to a prescribed deformation signal, the physical noise sources a real interferogram carries: a von-Kármán atmospheric phase screen (spatially correlated tropospheric delay), temporal and geometric decorrelation, a residual digital-elevation-model error term,

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Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_TailWatch; the studio is available at <https://tailwatch.fasl-work.com>.

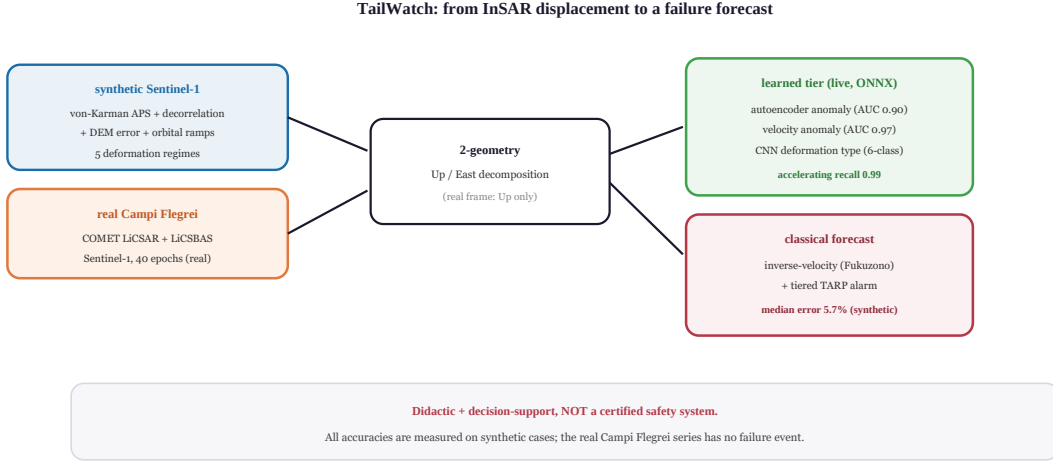


Fig. 1. The two-lane pipeline. A synthetic Sentinel-1 forward simulation (atmospheric phase screen, decorrelation, elevation-model error, orbital ramps) over five deformation regimes, and one real case (Campi Flegrei, COMET LiCSAR/LiCSBAS), feed an Up/East decomposition (two geometries on the synthetic lane; the single-geometry real frame gives the vertical component only), then a learned tier (anomaly detection and deformation-type classification, run live in the browser) and a classical inverse-velocity failure forecast with a tiered alarm. The accuracies shown are measured on synthetic data. The studio is a didactic and decision-support instrument, not a certified safety system.

and low-order orbital ramps. Five deformation regimes are simulated (stable, linear, accelerating, seasonal, and step), giving a controlled test set with exact ground truth for the learned tier and the forecast. The real lane is the Campi Flegrei caldera, a well-documented actively-deforming volcanic site, from the open COMET LiCSAR/LiCSBAS archive: descending frame 124D, forty epochs (of 67 available) spanning 2016–2018 over a 64×64 -pixel area, redistributed under the Copernicus free-and-open licence with the LiCSBAS attribution [7]. On the synthetic lane, a two-geometry decomposition combines line-of-sight measurements from ascending and descending viewing geometries into vertical (Up) and east-west (East) components, the SBAS-consistent step that turns a single ambiguous line-of-sight rate into an interpretable displacement (the full small-baseline network inversion is not implemented; see Section VI). The real frame has a single (descending) geometry: its East component is unavailable, and its vertical component is obtained from the line-of-sight up-vector under the assumption that the motion is vertical. On the real series the studio computes the line-of-sight velocity, the coherence, the cumulative displacement and its inverse velocity, and it runs the autoencoder anomaly detector and the classifier trained on synthetic scenes; with no ground truth on the real series, no accuracy is reported for those outputs.

III. THE LEARNED TIER

The learned tier, evaluated on held-out synthetic scenes (Fig. 2, Table I), does two jobs. First, *anomaly detection*: a velocity-based detector and a denoising convolutional autoencoder (trained only on stable ground, so a fault is a novelty) each flag pixels whose behaviour departs from the stable baseline. The velocity detector reaches area-under-curve 0.968 and the autoencoder 0.898, both strong, so the tier reliably answers “is anything moving?”. Second, *deformation-type classification*: a one-dimensional convolutional network

TABLE I
MEASURED ACCURACY OF EACH STAGE, ON SYNTHETIC DATA ONLY: THE LEARNED TIER ON HELD-OUT SYNTHETIC SCENES, THE FORECAST ON 40 SYNTHETIC ACCELERATING TRAJECTORIES AND 60 NON-ACCELERATING SYNTHETIC CONTROLS. THE REAL CAMPI FLEGREI SERIES HAS NO FAILURE EVENT AND NO GROUND TRUTH, SO NO ACCURACY IS MEASURED ON IT. THE LEARNED TIER IS STRONG AT ANOMALY DETECTION AND AT THE SAFETY-CRITICAL ACCELERATING CLASS; THE INVERSE-VELOCITY FORECAST DETECTS EVERY SYNTHETIC FAILURE WITH NO FALSE ALARMS.

Stage / metric	value
velocity anomaly detection (AUC)	0.968
autoencoder anomaly detection (AUC)	0.898
deformation-type classifier (macro-F1)	0.556
<i>accelerating</i> class recall	0.99
forecast detection rate (accelerating)	100%
forecast false-alarm rate	0%
median failure-time error	5.7%

labels each pixel’s time series as stable, linear, accelerating, seasonal, step, or decorrelated. Here the result is mixed. The overall six-class macro-F1 is a moderate 0.556, lowered by the hard distinctions: stable is confused with seasonal, and decorrelated pixels are recovered poorly. But the class that matters, *accelerating*, the signature of an impending failure, is recovered at 0.99 recall. A classifier that is mediocre at telling seasonal from stable but nearly perfect at catching acceleration makes the trade an early-warning tier should make, and the acceleration recall is therefore reported separately rather than folded into the macro-average.

IV. THE FAILURE FORECAST

The forecast is the decision the whole pipeline exists to support, and it uses the classical, interpretable inverse-velocity method: near failure a creeping slope’s velocity rises so that the inverse velocity falls approximately linearly to zero, and

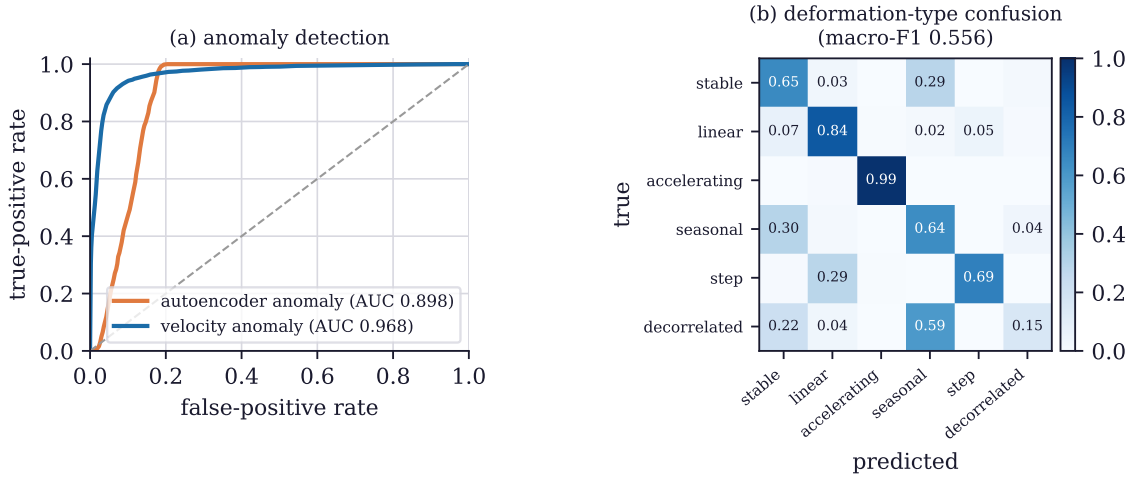


Fig. 2. The learned tier, on held-out synthetic scenes. **(a)** Anomaly detection: a velocity-based detector separates deforming from stable ground at area-under-curve 0.968, and a denoising-autoencoder novelty detector at 0.898. **(b)** The six-class deformation-type confusion matrix (macro-F1 0.556): the classifier is only moderate overall, confusing stable with seasonal and recovering decorrelated pixels poorly, but it identifies the safety-critical *accelerating* class at 0.99 recall, which is the class an early-warning system must not miss.

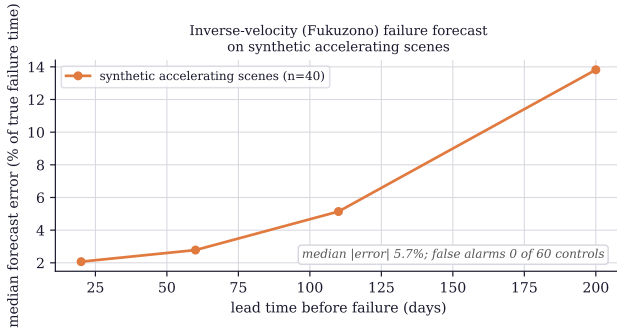


Fig. 3. The classical inverse-velocity (Fukuzono) failure forecast on the decomposed crest displacement of the 40 synthetic accelerating scenes. Median forecast error, as a percentage of the true failure time counted from the first epoch, against lead time. The error is about 2% close to failure and roughly 14% far ahead, the expected behaviour: as the failure approaches and the acceleration becomes unambiguous, the projection tightens. Every accelerating case is detected, and none of the 60 non-accelerating controls raises a false alarm. The real Campi Flegrei series has no failure event and is not scored.

extrapolating that line gives the failure time [8]. Applied to the decomposed crest displacement of 40 seeded synthetic accelerating scenes, with a two-trigger tiered alarm, the forecast (Fig. 3, Table I) detects every failure (100%) with no false alarm on 60 non-accelerating synthetic controls (stable, linear and seasonal, 20 each), and achieves a median failure-time error of 5.7% of the true failure time. Crucially, it behaves as a warning tool should: the error is about 14% when the forecast is made far ahead and tightens to roughly 2% close to failure, because the acceleration signature strengthens as the event nears. The Campi Flegrei series has no failure event in 2016–2018, so a failure-time error cannot be measured on it; the studio computes the inverse velocity of the real series, and any time-of-failure projection on it is illustrative only. The forecast contains no learned component: it is the inverse-velocity extrapolation of Fukuzono (1985) [8], and it comes

with an interpretable alarm a geotechnical engineer can audit.

V. DISCUSSION

The studio’s stages give complementary results on the synthetic test set. Anomaly detection is reliable (area-under-curve near 0.9–0.97 on held-out synthetic scenes), so “something is moving” is answered well. Fine-grained deformation-type classification is only moderate overall, and the confusion matrix (Fig. 2b) shows which distinctions are hard, but the one label that governs a safety decision is recovered almost perfectly. And the failure forecast, done with a classical inverse-velocity model rather than a learned one, is most precise close to failure, where it matters. On the real Campi Flegrei series, which has no failure event and no ground truth, none of these accuracies can be measured. The design lesson is the division of labour: learn the cheap, high-recall anomaly and acceleration flags, and forecast with a transparent physical model whose assumptions and alarm thresholds an engineer can inspect. The studio makes that whole chain reproducible and inspectable, with its accuracy measured on a controlled synthetic test set and the same chain applied to a real deforming site.

VI. SCOPE AND LIMITATIONS

TailWatch is a *didactic and decision-support* instrument, *not a certified safety system*; a real tailings-dam early-warning deployment requires site-specific calibration, redundant instrumentation, and a regulatory process that the studio does not replace. Most cases are the synthetic forward simulation; it is high-fidelity (real atmospheric, decorrelation and orbital noise models) but it is synthetic, and all learned-tier and forecast metrics are on that synthetic test set. There is one real case, Campi Flegrei: a single site, a volcanic (not tailings) one, observed in one (descending) geometry, so its vertical component rests on a vertical-motion assumption and its East component is unavailable; it has no failure event, so the forecast cannot be

scored on it, and the anomaly detector and classifier, trained on synthetic scenes, run on it without ground truth. The two-geometry decomposition assumes SBAS-consistent inputs; the full small-baseline network inversion from raw interferograms is not implemented. And the inverse-velocity forecast, like the method itself, assumes an accelerating-creep failure mode; brittle or non-creeping failures are outside its model. The studio labels the provenance of every case and states the same limits in its interface.

VII. CONCLUSION

TailWatch carries an InSAR displacement field to a failure forecast, with the accuracy of each step measured on synthetic data: strong anomaly detection (area-under-curve 0.90–0.97 on held-out synthetic scenes), a deformation-type classifier that is moderate overall (0.556 macro-F1) but nearly perfect on the safety-critical accelerating class (0.99 recall), and a classical inverse-velocity forecast that detects every one of 40 synthetic accelerating failures and predicts its time with a median error of 5.7%, tightening as the failure nears, with no false alarm on 60 controls. The same chain runs on the real Campi Flegrei Sentinel-1 series, which has no failure event, so no forecast accuracy is claimed for real data. The division of labour, learned high-recall flags plus a transparent physical forecast, together with an explicit scope, didactic and decision-support, mostly synthetic with one real case, defines the contribution: a reproducible, inspectable deformation-to-forecast studio rather than a black-box safety claim.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry the five synthetic regime traces and the real Campi Flegrei

cube and provenance, the learned-tier benchmark (anomaly ROC curves, the deformation-type confusion, the macro-F1 and area-under-curve values), and the forecast benchmark (detection rate, false-alarm counts on the controls and the per-lead-time error curve, all on synthetic scenes). The figure scripts under `manuscripts/insar-deformation/figures/` read those artifacts; the pipeline schematic is hand-authored. Because the synthetic generator and the forecast are seeded and deterministic, re-running reproduces Table I and Figs. 2–3; the real-case provenance and licence are recorded in `tw-real-cf.provenance.json`.

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